

1. (15 points) Answer each of the following questions by a simple “T” or “F” for true or false, and write your answers on *your answer book*.
- (a) (3 points) Based on Sklar’s theorem on copula function: for every joint distribution, under certain conditions, there is a unique copula function underlying the joint distribution, which, along with the marginal distributions from the joint distribution, would produce the same distribution as the original joint distribution.
  - (b) (3 points) For an equity tranche loss protection buyer, he or she would like that the underlying portfolio has higher correlation.
  - (c) (3 points) In US market, the legal claim that a bondholder has, when the bond defaults, is the market value of the bond just before default.
  - (d) (3 points) A tranche can be viewed as a call spread on the underlying portfolio loss distribution where the call strike prices are the subordination levels.
  - (e) (3 points) Using a Gaussian copula credit model, we can always find a correlation parameter called compound correlation which corresponds to each tranche traded spread as observed in the market.

**Solution:** (a) T; (b) F; (c) F; (d) T; (e) F.

2. (15 points) Suppose we have a zero coupon bond which matures at the end of the period  $[0, \Delta t]$ . If the bond does not default, you receive \$1 dollar, if it defaults, you receive a recovery rate  $R$ .
- (a) (10 points) Prove that this defaultable bond has approximately a present value as follows:
$$e^{-[r+(1-R)h]\Delta t},$$
where  $r$  is the short rate, and  $h$  is the hazard rate, both of which are assumed to be constant over this period.
  - (b) (5 points) Provide an intuitive explanation for this result, and state if this result could be extended for a general coupon bearing bond, or when interest rate and hazard rate are stochastic.

**Solution:**

- (a) Suppose the probability of survival over the period is  $p$  and default probability is  $q = 1 - p$ . The expected value is:

$$E(V) = [1 \times p + R \times q]e^{-r\Delta t}$$

$$1 \times p + R \times q = e^{-h\Delta t} + R(1 - e^{-h\Delta t}) \approx 1 - (1 - R)h\Delta t \approx e^{-(1-R)h\Delta t}.$$

Hence, we have

$$E(V) = 1 \times e^{-[r+(1-R)h]\Delta t}$$

- (b) This shows that we can value a *defaultable* cash flow *as if* it were default free, but adjust the discount factor by adding  $(1 - R)h$ , which can be interpreted as the mean loss or the expected loss or credit spread.

3. (20 points) In credit portfolio modeling, the correlation matrix could be interpreted as the asset return correlation matrix in the Gaussian copula model. We very often use a linear factor model to express this asset return correlation matrix.
- (a) (5 points) To come up with a factor model for pairwise constant correlation situation, and assume that correlation coefficient is  $\rho$ .
- (b) (10 points) To come up with a factor model for a two-correlation coefficients situation in which that intra correlation (within the same industry) is  $\rho_I$  and inter correlation (between two different industry) is  $\rho_O$ , and there are altogether  $K$  industries.
- (c) (5 points) to identify the number of independent risk factors in each of the above situations.

**Solution:**

- (a) For a specific company whose asset return is  $X_i$ , it can be expressed as follows:

$$X_i = \sqrt{\rho} X_M + \sqrt{1 - \rho} \epsilon_i$$

where  $X_M$  and  $\epsilon_i$  representing the systematic market factor and the idiosyncratic risk respectively, and they are independent and all follow a standard normal distributions.

- (b) For a specific credit name  $i$  in  $k$ 's th industry, where  $k = 1, 2, \dots, K$ , where is  $K$  is the total number of the industry sectors, we have the following

$$X_i^k = \sqrt{\rho_O} X_M + \sqrt{\rho_I - \rho_O} X_k + \sqrt{1 - \rho_I} \epsilon_i$$

where  $X_k$  is the factor representing the  $k$ th industry factor,  $X_M$  is still the market risk factor, and  $\epsilon_i$  is the indio-syncratic risk.

- (c) There are altogether  $K + 1$  independent factors.

4. (10 points) We have a random variable  $X$  whose distribution function is  $F(x)$  and whose density function is  $f(x)$ .

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- (a) (5 points) Prove that  $F(X)$  follows a uniform distribution over  $(0, 1]$ .
- (b) (5 points) Suppose that a random variable  $Y$  follows a distribution with the distribution function is  $F(x)$ . Show that  $\Phi^{-1}(F(X))$  follows the standard normal distribution where  $\Phi(x)$  is the distribution function for a standard Normal distribution.

**Solution:**

- (a) Suppose  $Y = F(X)$ .

$$F(y) = \Pr[Y \leq y] = \Pr[F(X) \leq y] = \Pr[X \leq F^{-1}(y)] = F(F^{-1}(y)) = y.$$

Hence,  $Y$  follows uniform distribution.

- (b) Suppose  $Y = \Phi^{-1}(F(X))$ , then we have

$$F(y) = \Pr[Y \leq y] = \Pr[\Phi^{-1}(F(X)) \leq y] = \Pr[F(X) \leq \Phi(y)] = \Phi(y),$$

so  $y$  follows the standard normal distribution, and the last step is based on the conclusion in the first question.

5. (20 points) (a) (5 points) We have a non-negative random random variable  $X$  whose distribution function is  $F(x)$  and whose density function is  $f(x)$ . Prove that

$$E(X) = \int_0^{\infty} (1 - F(x))dx$$

- (b) (5 points) We have a securitized tranche whose subordination levels are  $L_l$  and  $L_u$ , and the underlying portfolio loss distribution is described as a random variable  $X$  as in above. Express the loss of the tranche in terms of the portfolio loss  $X$ .
- (c) (5 points) Show that the loss for the tranche can be expressed as a call spread on the underlying loss distribution.
- (d) (5 points) Show that the expected loss for the tranche can be expressed as

$$\int_{L_l}^{L_u} (1 - F(x))dx$$

**Solution:**

- (a) For any non-negative random variable  $X$ , we can write

$$X = \int_0^{\infty} \mathbf{1}_{\{X > x\}} dx,$$

where  $\mathbf{1}_{\{X>x\}}$  is the indicator function that equals 1 when  $X > x$  and 0 otherwise. This identity holds because for a fixed outcome  $\omega$ , the integrand equals 1 for  $x \in [0, X(\omega))$  and 0 for  $x \geq X(\omega)$ , so the integral evaluates to  $X(\omega)$ .

Taking expectations of both sides:

$$E[X] = E \left[ \int_0^\infty \mathbf{1}_{\{X>x\}} dx \right].$$

Since  $\mathbf{1}_{\{X>x\}} \geq 0$ , we may apply Tonelli's theorem (the non-negative version of Fubini's theorem) to interchange the expectation and the integral:

$$E[X] = \int_0^\infty E[\mathbf{1}_{\{X>x\}}] dx = \int_0^\infty P(X > x) dx = \int_0^\infty [1 - F_X(x)] dx.$$

This completes the proof. Note that the result holds whether  $E[X]$  is finite or infinite.

(b) The payoff function for the tranche  $[K_L, K_U]$  is as follows:

$$L^T(t) = \begin{cases} 0, & L(t) < K_L \\ L(t) - K_L, & K_L \leq L(t) < K_U \\ K_U - K_L, & L(t) \geq K_U \end{cases}$$

(c) equivalently,  $L^T(t) = \max[L(t) - K_L, 0] - \max[L(t) - K_U, 0]$ .

Thus, we can view the tranche as a call spread on the total portfolio loss with strikes  $K_L$  and  $K_U$ .

(d) Next, we derive the expected values:

$$E[\max(L(t) - K_L, 0)] = \int_{K_L}^\infty (1 - F(x)) dx$$

Similarly,

$$E[\max(L(t) - K_U, 0)] = \int_{K_U}^\infty (1 - F(x)) dx$$

Therefore, the expected loss transfer is:

$$E[L^T(t)] = \int_{K_L}^{K_U} (1 - F(x)) dx$$

Here,  $1 - F(x)$  represents the *excess loss* distribution function. This shows that the expected tranche loss is simply the area below the excess loss distribution function bounded by the subordination levels<sup>1</sup>.

6. (10 points) In a one-factor Gaussian copula credit model, the correlation structure for standardized asset returns is as follow

$$X_i = \sqrt{\rho} X_M + \sqrt{1 - \rho} \epsilon_i$$

where  $X_i$ ,  $X_M$  are standardized asset returns, and the market asset return with mean zero and variance 1,  $\epsilon_i$ 's are non correlated noises with mean zero and variance 1,  $Cov(X_i, X_j) = \rho$  for  $i \neq j$ .

The survival time for each credit is linked to its asset return as follows

$$\tau_i = F_i^{-1}(N(X_i))$$

where  $F_i(t)$  is the survival time distribution function for credit  $i$ , and  $N()$  is the standard normal distribution function.

Conditional on the common market factor  $X_M$ , all credit survival times are independent to each other. Prove that the conditional default probability

$$q_i(t|X_M) = Pr[\tau_i < t|X_M]$$

can be linked to the non-conditional default probability

$$q_i(t) = Pr[\tau_i < t]$$

as follows:

$$q_i(t|X_M) = N\left[\frac{N^{-1}(q_i(t)) - \sqrt{\rho}X_M}{\sqrt{1 - \rho}}\right]$$

where  $N()$  is the standard normal distribution function, and  $q_i(t) = Pr[\tau_i < t]$ .

**Solution:**

$$\begin{aligned} q(t|X_M) &= Pr[\tau \leq t|X_M] \\ &= Pr[F^{-1}(\Phi(X)) \leq t|X_M] \\ &= Pr[X \leq \Phi^{-1}(F(t))|X_M] \\ &= Pr[\sqrt{\rho} \cdot X_M + \sqrt{1 - \rho} \cdot \epsilon \leq \Phi^{-1}(F(t))|X_M] \\ &= Pr\left[\epsilon \leq \frac{\Phi^{-1}(F(t)) - \sqrt{\rho} \cdot X_M}{\sqrt{1 - \rho}}|X_M\right] \\ &= \Phi\left(\frac{\Phi^{-1}(q(t)) - \sqrt{\rho} \cdot X_M}{\sqrt{1 - \rho}}|X_M\right) \end{aligned}$$

This formula gives the relationship between the unconditional default probability  $q(t)$  and the conditional default probability  $q(t|X_M)$ .

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7. (10 points) Provide a short summary about how various calculation methods for the credit portfolio loss distribution work: Monte Carlo, Conditional Normal approximation, and recursive method, and make comment on each of the methods in terms of calculation speed and accuracy.

**Solution:** Descriptions of the three methods:

- (a) Monte Carlo starts with simulating correlation asset return, this can be achieved by simulating  $n$  dimension correlated normal random variable. We can do this by first simulating independent  $n$  dimension normal random variables, and then apply a Cholesky decomposition to the given correlation matrix for the asset return, then multiply the independent normal random variables by this Cholesky decomposition. Finally, we use the relationship between the individual asset return and the its credit curve  $F(t_i) = N(X_i)$  for the survival time. With this information given, we can calculate the loss distribution over a period.
- (b) Conditional Normal: For the given parameter of the correlation assumption in the one parameter Gaussian copula model, then all credits become independent to each other. Then, the total loss is simply the sum of the independent individual losses, which can be approximated by a normal distribution under the central limit theorem. For the conditioning variable, which could be interpreted as the market index return, is integrated out using numerical integration based on Gaussian quadratures.
- (c) Conditional on the common factor, all individual credit losses become independent, then the portfolio loss can be taken as the sum of independent individual losses. Then, we can use the convolution method for the summation of the independent losses. The common factor can then be again integrated out by using Gaussian quadratures.
- (d) Monte Carlo method is easy to implement, but it could take a lot of runs to have an accurate convergent result. So it often suffers the speed challenge. This can be seen from the inverse of the credit curve, which is exponentially distributed. The inverse needs to have the survival time calculated under each run which very often involves to use the hazard rate to divide, which resulted in to having a large number of runs to converge. So, MC approach is often for quick prototype, and usually for one or a few runs of simulations.

Conditional normal approximation is fast, as it involves to obtaining just the first two moments. But it has the problem that conditional on the normal approximation, the loss distribution always has negative values while loss could be only positive. This can be compensated in the usage of the results. For example, we tend to change the equity tranche from  $[0, -3\%]$  to  $(-\infty, -3\%)$ .

The recursive method is the most accurate one as it involves no approximation. But we have to divide the whole range of total possible losses into many smaller bins,

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and allocate the probability of achieving a range of possible losses to the nearest bin with corresponding probability. But this approximation is easy for people to handle in practice.